

Improving Rare-Outcome Prediction With Separability-Aware Synthetic Health Data: An Initial Two-Dataset Validation Study

Oluwatoyin Joy Omole¹

¹Graduate Program in Computer Science, Universidade Federal de Minas Gerais, Belo Horizonte, Brazil
Author list, degrees, ORCIDs, and corresponding-author details to be confirmed

Abstract

Background: Synthetic data are often used to address rare outcomes by increasing minority-class sample size, but additional records may not improve prediction when classes overlap or when generation fails to preserve useful class-conditional structure.

Objective: This study asks when controlling class separability improves the held-out-real utility of synthetic health data beyond class-frequency correction and whether utility gains are accompanied by loss of minority coverage.

Methods: We developed SEPAWARE, a post-generation mechanism that selects records from an oversized candidate pool using realism and label-consistent geometric evidence while preserving a prespecified class count. Initial validation used a 5000-record sample of Brazil’s Vigitel obesity survey and a 334-record COVID-19 hospital-mortality cohort. Completed analyses include leakage-controlled 5-fold synthetic-to-real evaluation and a blocked 2×2 factorial experiment separating separability and anchor pressures. A strengthened augmentation analysis compares real-only training, random sampling, SMOTE, class weighting, conditional tabular generative adversarial networks, tabular variational autoencoders, and their SEPAWARE-selected variants. Evaluation includes discrimination, minority-specific performance, calibration, class-conditional coverage, and empirical privacy-risk proxies. A controlled simulation independently varies prevalence and overlap.

Results: In the synthetic-to-real experiment, Vigitel Macro-F1 increased from 0.459 with standard conditional tabular generative adversarial network data to 0.583 with SEPAWARE. Separability accounted for 96.00% of treatment-plus-error variation in the Vigitel factorial analysis ($P < .001$). COVID-19 results were less stable: standard generation achieved 0.463 and the fixed balanced SEPAWARE setting achieved 0.549 with wider uncertainty; separability accounted for 5.13% of variation and residual error for 81.53%. Across three repeated five-fold partitions in the strengthened augmentation analysis, SEPAWARE improved mean AUPRC over standard CTGAN by 0.015 on COVID-19 and 0.013 on Vigitel, but did not improve TVAE. Simple resampling or class weighting matched or exceeded generator-based approaches on most dataset–metric combinations. SEPAWARE CTGAN improved nearest-minority coverage while reducing minority diversity. Controlled SepAware-SMOTE gains were small and its overlap and prevalence interactions were not significant.

Conclusions: Initial evidence indicates that class-boundary control is distinct from class balancing and can materially improve synthetic-to-real transfer in some settings. The contrasting datasets also show why SepAware should not be treated as universally beneficial: apparent gains must be tested against uncertainty, calibration, minority coverage, and the possibility that selection merely removes difficult but valid cases.

Keywords: synthetic health data; tabular data; class imbalance; class overlap; rare outcomes; data augmentation; privacy; calibration

1 Introduction

Health datasets combine sensitive information, mixed numerical and categorical variables, missingness, and outcomes that are often uncommon but clinically consequential. Synthetic tabular data may support model development and collaboration when real records are scarce or difficult to share. Generation, however, does not guarantee privacy or downstream usefulness [1–3]. A generator can preserve aggregate distributions while weakening the class-conditional structure required to recognize a rare outcome. It can also increase apparent predictive performance by exaggerating relationships or selecting unusually easy cases.

Rare-outcome learning is commonly framed as a class-imbalance problem. Random sampling, SMOTE, class weighting, and conditional generation can increase the influence or number of minority observations, but additional records do not necessarily add information. The limiting factor may instead be class overlap: valid positive and negative observations can occupy similar regions of feature space. Class prevalence and class overlap are therefore different properties. Correcting the former does not necessarily repair the latter.

This distinction emerged from our qualification experiments. Conventional rebalancing, unconstrained CTGAN augmentation, and domain-guided candidate filtering produced limited or classifier-dependent gains. Nearest-neighbour and explainability analyses suggested that weak or unstable class geometry remained after augmentation. We therefore developed SEPAWARE, a post-generation selection mechanism that draws an oversized candidate pool and ranks records using proximity to the real training manifold and label-consistent geometric evidence. Selection retains a prespecified class count, allowing separability pressure to be studied without attributing every change to class ratio.

The central question is not whether SEPAWARE wins on every dataset. It is when control of class separability improves rare-outcome prediction, why the effect differs between datasets, and whether an improvement merely reflects removal of difficult but legitimate minority cases. The contrast already observed between Vigitel obesity and COVID-19 hospital mortality provides a useful initial test. Vigitel shows a large and stable synthetic-to-real gain under separability pressure, whereas the COVID-19 effect is smaller relative to fold variation. We treat this contrast as a boundary-condition signal rather than a result to conceal.

1.1 Research questions

We investigate three questions:

1. Is class frequency correction sufficient, or does class overlap limit the utility of synthetic augmentation?
2. Does separability-aware selection improve rare-outcome prediction beyond random sampling, SMOTE, class weighting, and unconstrained generation?
3. When performance improves, is the gain accompanied by acceptable calibration, minority coverage and diversity, and empirical privacy risk?

The first completed experiment tests mechanism in the synthetic-to-real regime. A leakage-controlled fixed-setting comparison is followed by a blocked 2×2 factorial analysis that separates separability and anchor pressures. The strengthened experiment evaluates practical augmentation using real-only training, conventional imbalance methods, CTGAN, TVAE, and SEPAWARE

variants on shared held-out real folds. It expands evaluation from Macro-F1 to AUPRC, minority-specific measures, AUROC, balanced accuracy, Brier score, calibration parameters, minority coverage and diversity, and privacy-risk proxies.

1.2 Contributions

This initial two-dataset study makes four contributions:

1. It frames class overlap as a distinct target from class imbalance and global fidelity in synthetic health-data augmentation.
2. It introduces a transparent post-generation selection mechanism whose effect can be separated from generator fitting and class-count correction.
3. It uses a factorial experiment to attribute the strong Vigitel synthetic-to-real change to separability pressure while reporting the COVID-19 null attribution with equal emphasis.
4. It tests the central rival explanation—that separability improves prediction by deleting difficult cases—through minority coverage, diversity, boundary, calibration, and privacy analyses in a strengthened augmentation protocol.

Because only two health datasets are presently available, we describe this work as initial validation. The experimental and manuscript structure is designed for additional datasets, but no claim of broad clinical generalization is made from the current evidence.

2 Related Work

2.1 Synthetic tabular health data

Synthetic microdata predate deep learning. Multiple imputation and probabilistic graphical models generate records from estimated conditional distributions and remain attractive because their assumptions can be inspected [4]. Copulas provide efficient dependence modelling but can struggle when discrete, continuous, multimodal, and rare-event variables interact nonlinearly [5].

Generative adversarial networks formulate generation as an adversarial game [6]. MedGAN adapted this idea to high-dimensional discrete patient records [7], and CTGAN addressed mixed-type tables through mode-specific normalization, conditional vectors, and training-by-sampling [8]. Variational autoencoders optimize a likelihood-related lower bound [9]; TVAE was evaluated alongside CTGAN in the same tabular-synthesis framework [8]. More recently, TabDDPM applied separate diffusion processes to numerical and categorical attributes and outperformed several GAN and VAE baselines on general tabular benchmarks [10]. The present strengthened analysis uses CTGAN and TVAE to test whether separability-aware selection transfers across generator families; diffusion remains an important future comparator rather than an assumed performance ceiling.

2.2 Evaluation beyond aggregate resemblance

Health-synthesis studies repeatedly conclude that no single metric establishes usefulness or safety and that evaluation should jointly cover resemblance, downstream utility, and privacy [1, 2, 11]. Recent comparisons report dataset- and metric-dependent rankings and a persistent fidelity/utility–privacy trade-off [12, 13]. A review of structured synthetic electronic health records also documents continuing problems with rare events, reproducibility, and privacy evaluation [14].

Univariate fidelity measures whether synthetic columns resemble their real counterparts, pairwise fidelity evaluates dependence structure, and multivariate distinguishability tests whether a classifier can separate real from synthetic records. None is sufficient for rare-outcome prediction. Aggregate scores can be dominated by majority-class records while the minority conditional distribution is poorly represented. Conversely, a selector can improve classification by producing easy prototypes rather than a faithful minority population.

Train-on-synthetic, test-on-real (TSTR) evaluation probes whether relations learned from synthetic data transfer to held-out real observations. Real-plus-synthetic-to-real evaluation instead addresses augmentation. These regimes answer different questions and must not be pooled. Neither is a privacy test or proof that inferential estimates are unbiased.

2.3 Class imbalance, overlap, and augmentation

Accuracy can remain high when minority recall is poor. Macro-F1 weights both classes equally, while AUPRC is often more informative than AUROC for rare positive outcomes [15]. Conventional methods include random sampling, SMOTE [16], cost-sensitive learning, and threshold adjustment [17]. These approaches correct frequency or decision cost, but they do not necessarily recover informative class-conditional structure.

A 2026 JMIR comparison across 13 small health datasets found that sampling with replacement could match more complex generative augmentation in several settings, suggesting that apparent gains may come primarily from increased sample size [18]. This makes random oversampling a required comparator and sharpens our question: does explicitly controlling generated class geometry add value beyond correcting sample counts?

Class imbalance and class overlap are related but distinct. A small minority class can remain geometrically well separated, whereas a moderately imbalanced outcome can be intrinsically difficult because valid observations occupy the same region of feature space. Adding examples cannot resolve irreducible overlap. Aggressively selecting distant examples may improve an empirical decision boundary while deleting difficult but legitimate cases. The relevant test is therefore whether improved discrimination is accompanied by acceptable calibration and preservation of minority coverage.

2.4 Privacy–utility tension

Synthetic data are not anonymous by construction. A high-capacity generator can reproduce or reveal information about its training records. Privacy should be examined through nearest-record, membership-inference, or attribute-inference analyses, or protected through a formal mechanism such as differential privacy [19, 20]. A recent scoping review found substantial heterogeneity in privacy and utility evaluation and warned that common assessments can underestimate risk [3]. LDP-GAN combines local differential privacy with medical-record synthesis [21], while a recent oncology framework evaluates candidate synthetic datasets across several quality dimensions before downstream use [22]. These studies support treating utility, fidelity, and privacy as separate outcomes rather than assuming that one implies the others.

2.5 Research gap

The literature leaves a practical gap at the intersection of rare outcomes, class-overlap diagnosis, transparent post-generation control, and multidataset validation. Generators optimize distribution learning, imbalance methods correct class frequency, and evaluation frameworks usually assess synthetic outputs after generation rather than steering class-conditional geometry. We test whether

separability is a distinct, attributable intervention and whether any utility improvement survives checks of calibration, minority structure, fidelity, and privacy risk.

3 Materials and Methods

3.1 Datasets

3.1.1 Vigitel health survey

Vigitel is Brazil’s telephone-based surveillance system for risk and protective factors for chronic non-communicable diseases [23]. The obesity experiments use a reproducible stratified sample of 5,000 rows drawn from a 2006–2023 extract, comprising 4,149 non-obese and 851 obese records (imbalance ratio approximately 4.9:1). The factorial experiment retains 15 predictors: age, sex, smoking, former smoking, physical inactivity, leisure-time physical activity, diabetes, hypertension, dyslipidaemia, dietary indicators, fruit/vegetable regularity, and schooling.

3.1.2 COVID-19 hospital mortality

The COVID-19 dataset contains 334 hospital records and 34 columns, including the binary target *obito* (death). The executed experiments report 251 survivors and 83 deaths, an imbalance ratio of approximately 3.0:1. Predictors include age, sex, alcohol and tobacco indicators, admission-reason groups, number of comorbidities, and specific conditions such as hypertension, heart failure, coronary disease, chronic respiratory disease, diabetes, chronic kidney disease, cancer, cirrhosis, and HIV.

The small sample size and the clinical heterogeneity of the death class are scientifically important rather than incidental. A synthetic-selection method can appear to improve separability simply by concentrating on an artificially homogeneous subset of deaths; results on this dataset must therefore be interpreted with wide uncertainty and assessed strictly on held-out real folds.

Table 1: Datasets used in the reported experiments. Counts are taken from executed repository outputs.

Dataset	n	Positive	Negative	Primary use
Vigitel obesity sample	5,000	851	4,149	SEPAWARE mechanism and augmentation experiments
COVID-19 mortality	334	83	251	SEPAWARE mechanism and augmentation experiments

3.2 Preprocessing and cross-validation protocol

Binary variables are normalised to $\{0, 1\}$; 1/2-coded survey fields are mapped deterministically. Within each of five stratified folds, numerical missing values are median-imputed, binary fields use the modal category, and continuous variables are scaled using statistics fitted only on that fold’s real training subset. CTGAN fitting, candidate scoring, selection, and classifier training all occur inside the fold; the corresponding held-out real fold is used only for evaluation.

Every predictive result reported in Section 4 uses this five-fold protocol. Within each factorial fold, a single CTGAN model generates a candidate pool five times the real training size, and the same pool is scored under all four factorial selection regimes described in Section 3.8. This pairing

removes irrelevant generator-to-generator variation between conditions, but it also means that the five fold scores are not independent laboratory replications, because the training folds overlap by 75% and the four within-fold conditions share a candidate pool. We return to this limitation in Section 5.6 and report a blocked statistical analysis designed to account for it.

We distinguish exploratory from confirmatory evidence throughout. All numerical claims in Sections 4 and 5 come from executed five-fold notebooks and their exported result tables; protocol-only, superseded single-split, or failed runs are not reported as evidence (see the reproducibility statement in A). Results from different evaluation regimes are not pooled: standalone synthetic substitution, real-plus-synthetic augmentation, and internal synthetic cross-validation answer different questions and are labelled separately below.

3.3 Evaluation regimes

Three regimes are kept distinct throughout:

Real→Real (TRTR)

train on four real folds and evaluate on the fifth real fold; this is the predictive reference.

Synthetic→Real (TSTR)

train only on synthetic records generated from a real training fold and test on the corresponding held-out real fold; this measures substitution utility.

Real+Synthetic→Real

append synthetic minority records to the real training fold and test on held-out real data; this measures augmentation, not synthetic substitution.

Macro-F1 is the endpoint of the completed qualification experiments because both outcome classes contribute equally to it. In the strengthened augmentation analysis, AUPRC is the prespecified primary endpoint. Secondary outcomes are Macro-F1, minority F1, minority precision and recall, AUROC, balanced accuracy, Brier score, and calibration intercept and slope. Values absent from an earlier export are not reconstructed after the fact.

3.4 Generators and predictive classifiers

CTGAN [8] is the generator in the completed mechanism experiments. The strengthened augmentation analysis adds TVAE [8] to test whether the selection mechanism transfers beyond a GAN. Standard-generator and SEPAWARE conditions share a candidate pool within a fold. Because SDV 1.17 does not implement native conditional sampling for TVAE, TVAE is fitted directly to minority-fold feature records and the outcome label is assigned by design; this is reported separately from CTGAN’s outcome-conditional sampling. Predictive models are logistic regression, random forest, CatBoost [24], and, in earlier analyses only, XGBoost [25] and TabPFN [26].

3.5 SepAware selection framework

For real training data $\mathcal{D}_R$ and candidate pool $\mathcal{C}$, each candidate $(\tilde{x}, \tilde{y})$ receives a composite score

$$S(\tilde{x}, \tilde{y}) = R(\tilde{x}) + \lambda_{\text{sep}} \text{Sep}(\tilde{x}, \tilde{y}) + \lambda_{\text{anchor}} A(\tilde{x}, \tilde{y}). \quad (1)$$

R is one minus a robustly normalised nearest-real distance. Sep averages a nearest-hit/nearest-miss margin and the assigned-label probability from a real-trained random forest. A averages compatibility with predeclared class-conditional anchor patterns. Candidates are ranked within target

class and selected to reproduce the real training-fold class counts, so improved TSTR performance cannot be attributed to a changed class ratio.

The fixed- λ grid uses $\lambda_{\text{sep}} \in \{0, 0.25, 0.5, 1\}$ and $\lambda_{\text{anchor}} \in \{0, 0.25, 0.5\}$; because the best configuration on this grid is chosen using the same outcomes later used for reporting, grid results are treated as exploratory (Section 4.2). The subsequent factorial experiment instead predeclares two levels per factor, $\lambda_{\text{sep}} \in \{0, 1\}$ and $\lambda_{\text{anchor}} \in \{0, 0.5\}$, so that its conclusions do not depend on post-hoc selection.

Figure 1 summarises the pipeline. CTGAN produces candidates but does not itself optimise the three selection criteria; a separate selector scores and ranks candidates within class and returns the requested class counts. Separating generation from selection makes the intervention auditable and allows component-wise ablation, which is the basis of the factorial design in Section 3.8. The design rules out one simple alternative explanation – that a TSTR change is caused by a different class ratio – because class counts are held fixed across all compared conditions. It does not rule out harder alternatives: ranking can select unusually easy prototypes, discard boundary cases, or amplify a classifier-specific shortcut. These possibilities motivate the confirmatory analyses proposed in Section 6.

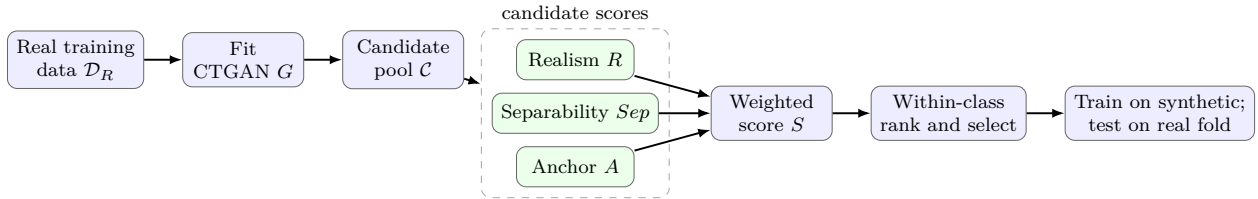


Figure 1: SEPAWARE post-generation selection pipeline. Every operation is fitted or computed using the real training portion of a cross-validation fold; the held-out real fold is used only for evaluation.

3.6 Fidelity, separability, and privacy proxies

Fidelity combines numerical-distribution similarity, categorical-proportion similarity, and correlation preservation into a single composite score. Separability is reported using cosine silhouette, a one-nearest-neighbour label-agreement index (SI), and hypothesis margin (HM). These are descriptive geometric measures: higher separation is not automatically clinically desirable, and we do not treat it as such (Section 5).

The strengthened experiment measures held-out-real minority coverage as the mean distance from each positive test record to its nearest synthetic minority record after training-fold standardization. Minority diversity is the mean pairwise standardized distance among up to 1000 selected minority records. Exact overlap is the proportion of synthetic records with zero distance to a real training record. The membership proxy compares each real training and held-out record’s distance to the nearest synthetic record and reports attack AUROC. These are empirical diagnostics, not evidence of anonymity or a formal guarantee.

3.7 Domain-anchored causal plausibility filtering

DACPF generates minority-class CTGAN candidates and ranks their consistency with real minority distributions for domain anchors. The completed COVID-19 experiment compares no augmentation, standard CTGAN augmentation, age filtering, disease-anchor filtering, and combined filtering,

evaluated as real-plus-synthetic augmentation (Real+Synthetic→Real) rather than standalone synthetic substitution. We emphasise that DACPF is a plausibility filter, not causal discovery or effect estimation: “causal” describes the clinical origin of the anchor hypotheses, not a claim that the filtering step identifies a causal effect.

3.8 Replicated 2^2 factorial experiment

Let $A \in \{-1, +1\}$ encode separability pressure and $B \in \{-1, +1\}$ anchor pressure. The fold-level response is the classifier-averaged TSTR Macro-F1,

$$y_{ijk} = \mu + q_A A_i + q_B B_j + q_{AB} A_i B_j + \gamma_k + \varepsilon_{ijk}, \quad (2)$$

where γ_k is a fixed fold block. The high-minus-low factorial effects are $2q_A$, $2q_B$, and $2q_{AB}$. Type-II ANOVA partitions sums of squares among factors, interaction, fold, and residual error; the primary variation percentages exclude the fold block from the denominator so that factor contributions are compared against experimental error rather than against total variation. An unblocked $2^2 r$ decomposition that treats folds as replications is also reported in the supplementary material for comparison, but Equation 2 is the more defensible model for paired cross-validation conditions and is used for all primary conclusions.

Both raw and log-transformed responses are analysed. Residual normality is assessed using Shapiro–Wilk and Jarque–Bera tests; constant variance is assessed with median-centred Levene (Brown–Forsythe), Fligner–Killeen, and Breusch–Pagan tests; Durbin–Watson and lag-one Ljung–Box statistics diagnose ordering-related serial patterns but cannot themselves establish independence. As noted in Section 3.2, five-fold training sets overlap by 75% and the four conditions within a fold are paired, so inference from this design is treated as exploratory rather than as a formal, independently replicated hypothesis test (Section 5.6).

3.9 Strengthened augmentation experiment

The augmentation experiment uses three repeated five-fold stratified partitions and compares real-only training, random oversampling, random undersampling, SMOTE, class-weighted training, standard CTGAN, SEPAWARE CTGAN, standard TVAE, and SEPAWARE TVAE. Synthetic conditions add positive candidates until the real training fold is balanced; the held-out fold retains its natural prevalence. All missing-value imputation is fitted on the real training fold before resampling or generation. The SEPAWARE augmentation score uses realism plus separability with a fixed unit separability weight; no held-out outcome is used to tune the setting.

Logistic regression supplies a linear and calibration-sensitive reference, random forest supplies a nonlinear ensemble, and CatBoost supplies a strong tabular learner. Hyperparameters and the 0.5 classification threshold are held fixed across methods. Fold-level results are retained so generator comparisons can be paired within dataset, repeat, fold, and classifier. The current manuscript run uses three independently shuffled five-fold partitions with independent fold-specific generator seeds. These repetitions improve stability assessment but remain internal repetitions of the same two dataset samples rather than external validation.

3.10 Controlled prevalence-by-overlap experiment

A separate mechanism experiment varies class prevalence and overlap independently and is not counted as a third health dataset. We generated 2000-record, 15-feature binary-classification problems with 8 informative features, 3 redundant features, 2 clusters per class, and 2% label noise.

Prevalence was set to 5%, 15%, or 30%, and class separation to 0.4, 1.0, or 2.0, corresponding respectively to high, moderate, and low overlap. Each of the 9 cells used 10 independent data seeds and a stratified 70/30 train-test split.

The comparison includes real-only training, random oversampling, SMOTE, and a SepAware-SMOTE condition. For the latter, SMOTE generates five times the number of required minority candidates; the selector ranks them by nearest-real realism, nearest-hit/nearest-miss margin, and random-forest label confidence, then selects enough records to balance the training data. Logistic regression and random forest are evaluated with AUPRC, Macro-F1, and minority F1. A heteroscedasticity-robust linear model tests method-by-overlap and method-by-prevalence interactions.

3.11 Ethical considerations

The Vigitel analysis uses a secondary survey extract and the COVID-19 analysis uses a deidentified institutional dataset supplied for the qualification research. The manuscript does not infer a new ethics approval number or consent determination. The authors must insert the applicable institutional review board or ethics committee name, protocol identifier, consent or waiver determination, and any access restrictions before submission.

3.12 Reproducibility

The strengthened experiment was executed with Python 3.12, NumPy 1.26.4, pandas 2.2.2, scikit-learn 1.6.1, SDV 1.17.2, CatBoost 1.2.10, XGBoost 3.2.0, imbalanced-learn 0.12.3, and statsmodels 0.14.2. A project-local environment pins NumPy 1.26.4 because the available compiled scientific extensions were incompatible with NumPy 2. Full software-stack notes, executed-notebook identifiers, and artefact-status decisions are given in the Code and data availability statement and in [A](#).

4 Results

4.1 Rebalancing and unconstrained augmentation are insufficient

The project began from the conventional explanation for poor rare-outcome prediction: that the minority class simply supplies too few examples. We first examined ordinary rebalancing, CTGAN minority augmentation, and increasingly targeted candidate filters. The common result across these attempts was not that augmentation never helped, but that additional minority records did not produce a sufficiently large and consistent gain to explain the problem by class frequency alone. Some of these early notebooks were single-split or diagnostic studies and are treated as hypothesis-generating rather than confirmatory evidence.

The leakage-controlled DACPF rerun provides the strongest five-fold evidence from this stage (Table 2). Standard CTGAN augmentation improves mean Macro-F1 relative to real-only training for both classifiers available in the local environment. Adding an age filter, or a combined age-and-disease filter, does not improve on standard CTGAN. Disease-anchor filtering performs best, adding 0.0508 over standard CTGAN for CatBoost but only 0.0084 for XGBoost. Domain targeting can therefore help, but its effect depends on both the anchor and the classifier, and adding more constraints is not monotonically beneficial.

This stage ruled out a simple monotonic story in which balancing classes, generating more synthetic cases, or imposing more domain constraints necessarily improves prediction. We then

Table 2: Five-fold COVID-19 augmentation Macro-F1. These are Real+Synthetic→Real results and are not directly comparable with the standalone-synthetic TSTR results in Section 4.2.

Condition	CatBoost	XGBoost
Real only	0.4740	0.4759
Real + standard CTGAN	0.5262	0.5255
Real + age-filtered CTGAN	0.5257	0.4983
Real + disease-filtered CTGAN	0.5771	0.5339
Real + combined DACPF	0.5256	0.5204

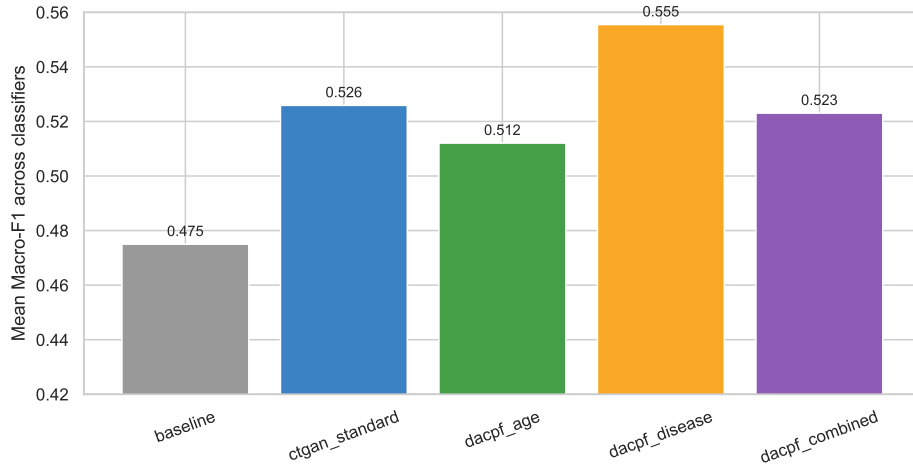


Figure 2: COVID-19 DACPF Macro-F1 averaged across CatBoost and XGBoost. The classifier-specific values in Table 2 show why this average should not be interpreted as a universal anchor effect.

used explainability and neighbourhood analyses to examine diversity, class-wise recall, feature-importance shifts, silhouette, nearest-neighbour label agreement, and hypothesis margin. Because some of these analyses used representative folds or earlier configurations, they establish the diagnostic rationale for what follows rather than a further confirmatory result. Their implication was that weak or unstable class geometry remained after augmentation, which led to the working hypothesis that class overlap, rather than imbalance alone, was limiting synthetic-to-real transfer.

4.2 Separability as an explicit generation objective

SEPAWARE is the intervention implied by that diagnosis. It does not change the target class counts; instead it asks whether records selected for label-consistent geometry transfer better to held-out real cases. The fixed- λ experiment fits the complete pipeline independently in five stratified folds. Table 3 reports classifier-averaged fold means and 95% t intervals. On COVID-19, the balanced setting reaches 0.5492 Macro-F1, but its interval is wide. On Vigitel, both the varied grid and the fixed high-separability setting reach approximately 0.58 and outperform ordinary CTGAN in all ten matched fold-classifier comparisons. The varied-grid result remains exploratory because the configuration was selected using the same folds used for reporting.

Table 3: Five-fold TSTR Macro-F1. Means and 95% intervals are computed from five classifier-averaged fold scores.

Dataset	Training data	Mean	95% half-width
COVID	Real	0.5069	0.0690
	Standard CTGAN	0.4627	0.0407
	SEPAWARE varied grid	0.5350	0.0649
	Fixed balanced (1.0, 0.5)	0.5492	0.0959
Vigitel	Real	0.4672	0.0088
	Standard CTGAN	0.4588	0.0107
	SEPAWARE varied grid	0.5839	0.0225
	Fixed balanced (1.0, 0.5)	0.5834	0.0216

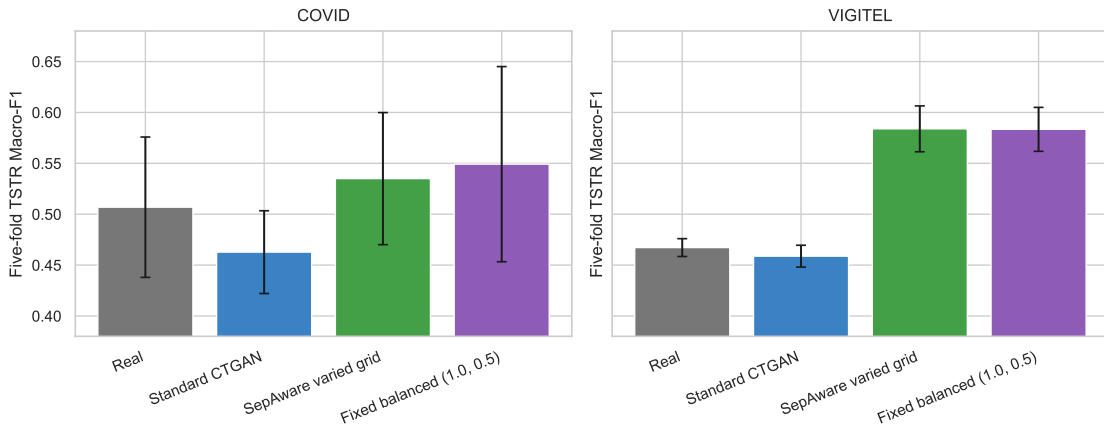


Figure 3: Five-fold TSTR Macro-F1 for the principal fixed- λ comparison. Error bars are 95% t intervals over the five classifier-averaged fold scores.

Table 4 reports the highest mean selection scores from the same run. The repository composite fidelity score varies little across configurations, whereas SI and HM respond strongly to separability pressure. Figure 4 shows the same pattern across the full grid: global fidelity is comparatively stable while separability changes substantially. This is evidence that the selector controls a property not captured by the composite fidelity score; it is not evidence that the selected records are clinically representative.

Table 4: Top executed SEP-AWARE grid configurations. “Quality” is the repository composite fidelity score.

Dataset	$(\lambda_{sep}, \lambda_{anchor})$	Quality	SI	HM	Selection
COVID	(1.0, 0.0)	0.956	0.779	0.897	2.154
COVID	(1.0, 0.25)	0.961	0.760	0.854	2.095
COVID	(1.0, 0.5)	0.965	0.754	0.830	2.067
Vigitel	(1.0, 0.5)	0.967	0.812	0.829	2.124
Vigitel	(1.0, 0.25)	0.967	0.811	0.822	2.116
Vigitel	(1.0, 0.0)	0.966	0.811	0.822	2.116

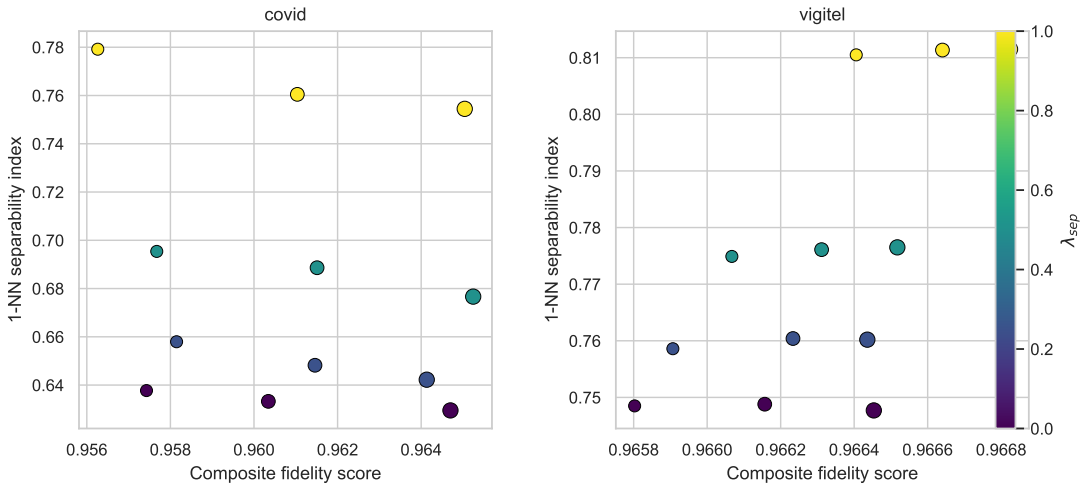


Figure 4: Five-fold mean fidelity–separability trade-off across executed λ settings. Colour represents separability weight; marker size represents anchor weight.

4.3 Attributing the gain: replicated factorial experiment

The fixed- λ results raise a sharper question: was the apparent gain produced by separability pressure, anchor pressure, or their interaction? The replicated 2^2 experiment (Section 3.8) answers this component-level question under four predeclared settings. Figure 5 displays all 40 fold responses. COVID-19 profiles cross and show substantial fold-to-fold variation; Vigitel shows a stable step increase when separability is enabled.

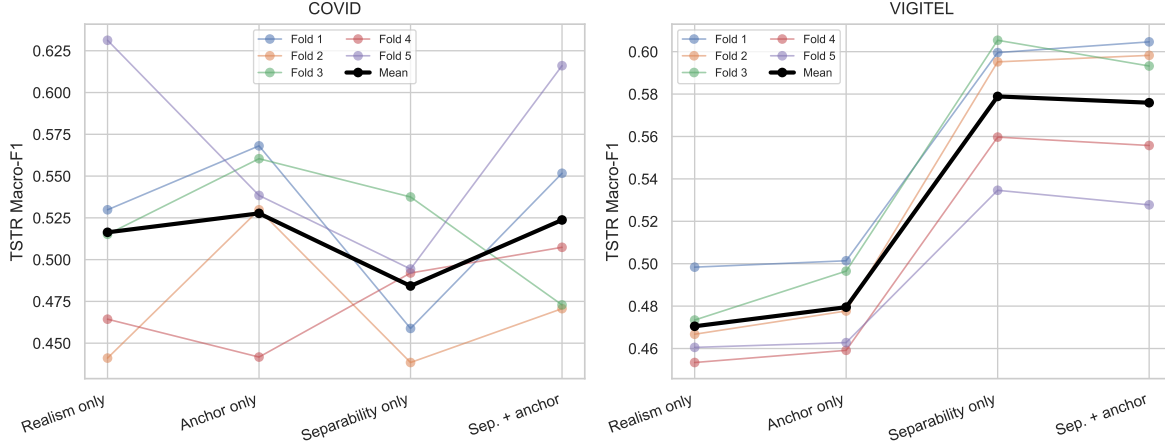


Figure 5: Fold-level TSTR Macro-F1 in the replicated 2^2 SEPAWARE experiment. Black lines show condition means.

Table 5: Mean TSTR Macro-F1 by predeclared factorial condition. Values are means of the five exported fold responses.

Dataset	Realism only	Anchor only	Separability only	Both
COVID	0.516	0.528	0.484	0.524
Vigitel	0.470	0.480	0.579	0.576

The condition means in Table 5 make the contrast explicit. In Vigitel, enabling separability raises the mean from approximately 0.47–0.48 to 0.576–0.579, while adding anchors makes little difference. COVID-19 shows no corresponding ordering: separability-only is the *lowest* condition mean, and the combined condition returns approximately to realism-only performance.

The blocked raw-scale ANOVA in Table 6 is the primary component analysis. For COVID-19, none of the treatment effects is significant, and residual error accounts for 81.53% of treatment-plus-error variation after separating out the fold block. For Vigitel, separability accounts for 96.00% of that variation and is highly significant ($p = 5.03 \times 10^{-10}$), while anchor and interaction contributions are negligible. The log-scale model yields the same substantive conclusion.

Table 6: Blocked factorial ANOVA: percentage of experimental variation and p -values on raw Macro-F1. Percentages exclude fold sum of squares from the denominator.

Dataset	Source	Variation (%)	p
COVID	Separability	5.13	0.4019
	Anchor	10.21	0.2439
	Interaction	3.13	0.5103
	Error	81.53	–
Vigitel	Separability	96.00	5.03×10^{-10}
	Anchor	0.08	0.6069
	Interaction	0.33	0.3154
	Error	3.59	–

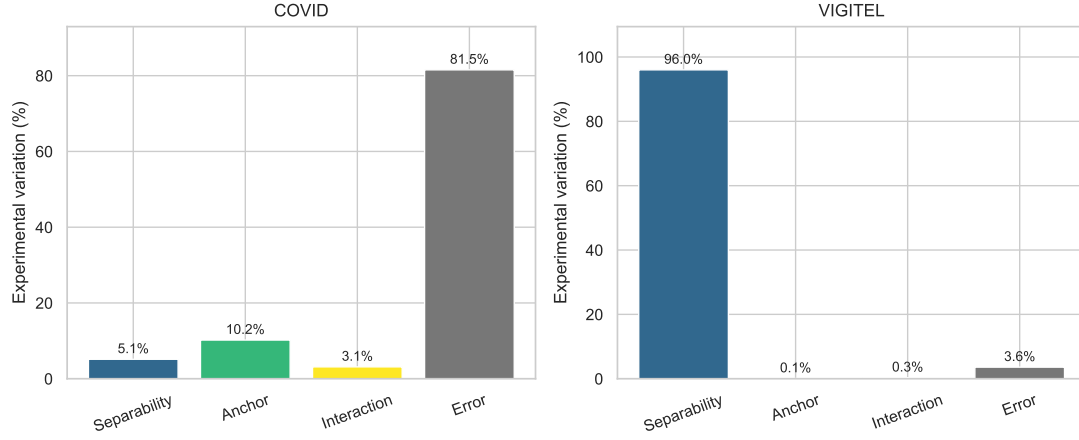


Figure 6: Variation allocation for the blocked raw-scale factorial models.

Table 7: Residual assumption diagnostics for the blocked factorial models.

Dataset	Scale	Shapiro	B-F	Fligner	B-P	Assessment
COVID	raw	.2446	.8409	.8394	.4415	no detected violation
COVID	log	.2031	.8173	.8050	.6435	no detected violation
Vigitel	raw	.8564	.8413	.7706	.0186	variance flag
Vigitel	log	.8625	.7463	.6834	.0237	variance flag

Normality tests do not reject for either dataset. Vigitel’s groupwise robust variance tests (Brown–Forsythe, Fligner–Killeen) do not reject equal variance, but Breusch–Pagan does, and the log transform does not resolve this conflict. More importantly, independence is not established, because the cross-validation training sets overlap and the four conditions within each fold share a candidate pool. The very small Vigitel residual should therefore be confirmed across independent generator and partition seeds before it is treated as a stable population effect (Section 5.6).

4.4 Strengthened augmentation comparison

The practical augmentation analysis gives a more restrained result than the synthetic-to-real factorial experiment (Table 8). Across three repeated partitions and three classifiers, SEPAWARE CTGAN produced the highest COVID-19 Macro-F1 (0.541), followed by random oversampling (0.535) and class weighting (0.533). Random oversampling produced the highest Vigitel Macro-F1 (0.575), approximately equal to class weighting (0.574). Vigitel AUPRC was highest for real-only training (0.288) and class weighting (0.287); COVID-19 AUPRC was similar for random oversampling (0.357), SEPAWARE CTGAN (0.354), and real-only training (0.354).

Within-generator paired comparisons were directionally favorable for CTGAN but not TVAE. Relative to standard CTGAN, SEPAWARE increased mean AUPRC by 0.015 on COVID-19 and 0.013 on Vigitel; corresponding Macro-F1 differences were 0.033 and 0.019. Relative to standard TVAE, SEPAWARE changed AUPRC by -0.015 on COVID-19 and -0.001 on Vigitel and did not improve Macro-F1. The post-generation mechanism therefore transferred across the two datasets for CTGAN in this run, but not across generator families.

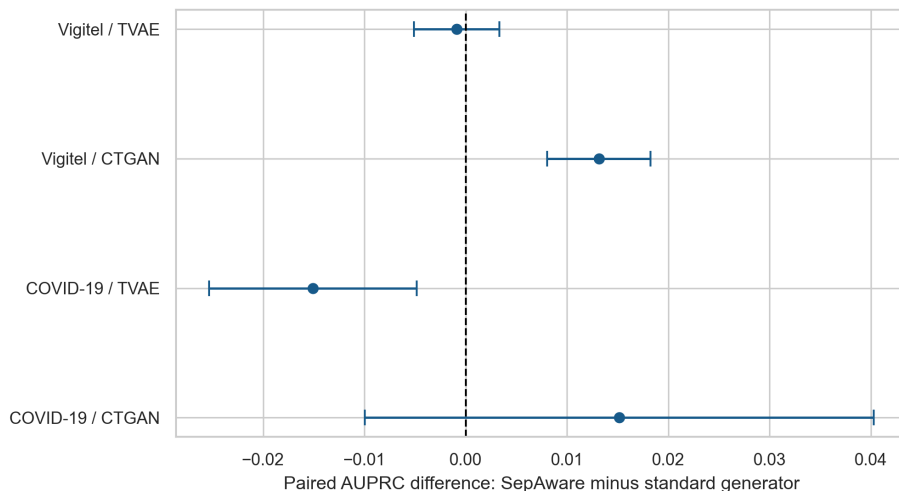


Figure 7: Paired AUPRC differences between SEPAWARE and the corresponding standard generator across shared fold-classifier observations. Error bars are conventional 95% t intervals and are descriptive because cross-validation folds are not independent replications.

The structural diagnostics reveal the trade-off behind the CTGAN improvement. SEPAWARE reduced the mean distance from held-out real minority records to their closest synthetic minority record (COVID-19, 4.30 vs 4.44; Vigitel, 1.71 vs 1.99), indicating improved nearest-record coverage. At the same time, mean within-minority diversity declined (COVID-19, 7.17 vs 11.12; Vigitel, 4.33 vs 5.55). No exact standardized matches were observed. Membership-proxy AUROC remained close to chance, ranging from 0.50 to 0.52, but this limited proxy is not a privacy guarantee.

Table 8: Held-out-real augmentation performance averaged across 3 repeated partitions, 5 folds, and 3 classifiers. AUPRC was the prespecified primary endpoint of the strengthened analysis.

Dataset	Method	AUPRC	Macro-F1	Minority F1	Brier
COVID-19	Real only	0.354	0.498	0.159	0.196
	Random oversampling	0.357	0.535	0.336	0.230
	Random undersampling	0.337	0.504	0.379	0.279
	SMOTE	0.323	0.508	0.237	0.219
	Class weighting	0.352	0.533	0.334	0.233
	Standard CTGAN	0.339	0.508	0.286	0.236
	SepAware CTGAN	0.354	0.541	0.335	0.231
	Standard TVAE	0.347	0.524	0.237	0.211
	SepAware TVAE	0.332	0.495	0.183	0.213
Vigitel	Real only	0.288	0.463	0.021	0.134
	Random oversampling	0.284	0.575	0.351	0.202
	Random undersampling	0.279	0.552	0.365	0.230
	SMOTE	0.237	0.506	0.174	0.175
	Class weighting	0.287	0.574	0.346	0.200
	Standard CTGAN	0.237	0.508	0.175	0.176
	SepAware CTGAN	0.250	0.527	0.215	0.176
	Standard TVAE	0.253	0.493	0.107	0.154
	SepAware TVAE	0.252	0.487	0.093	0.153

TVAE minority diversity was especially low and SEPAWARE did not improve its predictive utility, consistent with a class-conditional mode-concentration failure.

These results do not support the claim that increasing synthetic separability is generally better. They support a narrower conclusion: within CTGAN candidate pools, separability-aware selection can yield a small average improvement and closer minority coverage, but the gain may come with reduced diversity and can be dominated by simpler frequency-correction methods.

4.5 Controlled prevalence and overlap

The controlled experiment confirms that prevalence and overlap exert distinct effects on rare-outcome performance. AUPRC increases sharply as separation rises and as the positive class becomes more prevalent. For logistic regression, SepAware-SMOTE exceeds ordinary SMOTE in all 9 prevalence–overlap cells, but the mean differences are small (0.001–0.024). The largest gains occur at 5% prevalence with moderate or low overlap and at 15% prevalence with high or moderate overlap; the benefit approaches zero at 30% prevalence with low overlap (Figure 8).

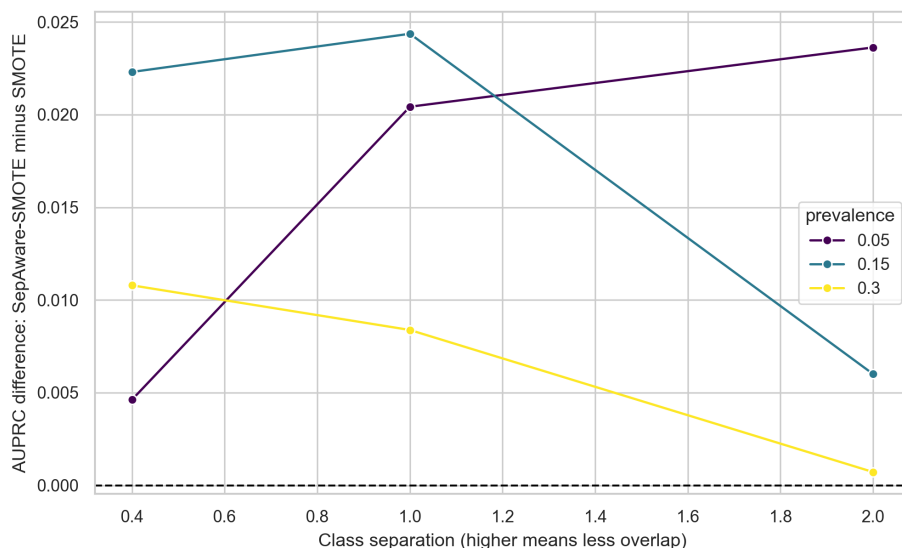


Figure 8: Controlled mechanism experiment for logistic regression. Each point is the mean AUPRC difference between SepAware-selected and ordinary SMOTE across 10 independent datasets. Higher class separation indicates less overlap.

In the pooled robust regression including both classifiers, neither the SepAware-SMOTE main contrast nor its interactions with overlap or prevalence were statistically distinguishable from zero (all $P > .91$). The simulation therefore supports the distinction between prevalence and overlap but does not establish a stable region in which separability-aware SMOTE selection is superior. It provides no evidence for imposing the hypothesized inverted-U relationship on the observed health datasets.

4.6 Summary of the empirical evidence

The experiments form a coherent progression. Rebalancing and unconstrained augmentation did not make utility reliably high; domain filters sometimes helped but did not behave monotonically; diagnostic analyses implicated class overlap; SEPAWARE then improved Vigitel TSTR performance

while measured composite fidelity remained broadly stable; and the factorial study attributed that gain to separability rather than to anchors. The COVID-19 null result is equally informative: explicit separability pressure is not sufficient when the positive class is small, heterogeneous, or weakly represented.

The strengthened augmentation results qualify that progression. CTGAN selection improves average AUPRC modestly on both datasets, but random oversampling is highly competitive, TVAE selection does not improve performance, and the CTGAN coverage gain accompanies lower minority diversity. The evidence therefore does not show that SEPAWARE recovers clinically meaningful signal. An equally consistent explanation is that it concentrates on easier observations. We treat these as central boundary conditions rather than incidental caveats.

5 Discussion

5.1 Principal findings

The study produces two complementary findings. First, the completed synthetic-to-real factorial experiment shows that separability is an attributable, utility-relevant property in Vigitel: enabling separability pressure produces an approximately 0.10 Macro-F1 step, while anchor pressure contributes little. The corresponding COVID-19 attribution is not established because residual fold variation dominates. Second, the strengthened augmentation analysis shows that this large substitution result should not be interpreted as a general augmentation advantage. SEPAWARE improves CTGAN AUPRC modestly on both datasets, but random oversampling is highly competitive and SEPAWARE does not improve TVAE.

This distinction matters. TSTR asks whether a synthetic dataset can substitute for real training data; augmentation asks whether synthetic minority records add value when the original training records remain available. A method may improve the former without being the best practical solution for the latter. For Vigitel, real-only AUPRC was 0.288, random oversampling achieved 0.284, and SEPAWARE CTGAN achieved 0.250. Its advantage was relative to standard CTGAN, not relative to every alternative. For COVID-19, random oversampling achieved the highest mean AUPRC (0.357), while SEPAWARE CTGAN and real-only training both achieved 0.354.

5.2 Comparison with prior work

The competitiveness of random oversampling agrees with a 2026 JMIR study in which sampling with replacement matched substantially more complex augmentation in small-data settings [18]. Our results extend that caution to rare outcomes: greater model complexity does not ensure that augmentation adds information. The generator and its sampling mechanism can matter as much as the downstream selector.

Contemporary evaluation frameworks emphasize that fidelity, utility, and privacy can move in different directions [12, 13]. We observe the same pattern within the minority class. SEPAWARE CTGAN improves nearest-minority coverage while reducing pairwise minority diversity. A single aggregate fidelity score would not expose that trade-off. The finding also supports calls for explicit privacy and utility threat models rather than duplicate checks alone [3].

5.3 When separability helps

The Vigitel–COVID contrast suggests that separability pressure interacts with the structure and stability of the outcome class. Vigitel provides a larger sample and a more repeatable factorial

response. COVID-19 contains only 83 deaths and a heterogeneous set of admission reasons and comorbidities. In that setting, the same scalar margin can be unstable or can compress distinct mortality patterns. The repeated CTGAN augmentation results nevertheless show small positive AUPRC differences for both datasets, suggesting that the selector can improve a candidate pool even when the factorial attribution is weak. This pattern remains exploratory until additional datasets and external or temporal validation test its portability.

The negative TVAE result is equally important. SDV’s TVAE does not support native outcome-conditional sampling, so the experiment fitted a minority-conditional feature model. The resulting synthetic minorities had very low diversity, especially for Vigitel. Post-selection could not recover information absent from the candidate pool and sometimes reduced performance further. SEPAWARE is therefore not generator agnostic merely because its scoring rule can technically be applied to another generator.

5.4 The easier-case explanation

Higher separability can represent clearer class-conditional signal, but it can also represent deletion of difficult cases. The strengthened diagnostics support both interpretations. Smaller held-out-minority coverage distance is favorable: selected CTGAN records lie closer to real positive test cases. Lower within-minority diversity is unfavorable: the selector concentrates the candidate distribution. Because no externally validated phenotype labels are available, we cannot determine whether the lost variation is noise, redundant spread, or clinically meaningful heterogeneity.

This trade-off changes the decision rule for the method. A Macro-F1 or AUPRC gain alone is insufficient. A favorable use case requires improvement relative to the corresponding standard generator without material loss of calibration, minority coverage, or clinically meaningful diversity. Boundary-stratified error analysis and expert-defined minority phenotypes are needed to determine how much separability is too much.

5.5 Privacy implications

No exact matches were observed and the membership proxy remained near chance. These are reassuring diagnostics but not evidence of anonymity. SEPAWARE explicitly prefers records near the real manifold, so the same mechanism that improves utility could increase disclosure risk under a stronger attacker. Membership and attribute inference under explicit attacker knowledge, rarity-stratified analyses, and formal privacy mechanisms remain necessary before any synthetic records are released.

5.6 Statistical interpretation

The factorial conditions share candidate pools within folds and cross-validation training sets overlap. Blocking captures the paired design, but classical P values remain conditional on one partition, generator configuration, and dataset sample. The strengthened augmentation analysis uses three repeated five-fold partitions with three classifiers; its 45 fold-classifier observations still share source records and do not constitute 45 independent clinical samples. The reported paired intervals are therefore descriptive. Independent datasets and external or temporal validation are required for confirmatory inference.

The appropriate conclusion is therefore conditional. Separability-aware selection controls a property not captured by class balance or global fidelity and can improve CTGAN relative to its unselected candidate pool. It is not uniformly beneficial across generators, it does not consistently

beat simple oversampling, and its coverage gains may be purchased with reduced diversity. Those boundary conditions are the scientific contribution of the current two-dataset study.

6 Limitations

Only two health datasets were available. They differ substantially in size and outcome context, which makes the contrast informative, but two datasets cannot identify a general relationship between intrinsic overlap and SepAware benefit. The current paper should therefore be read as an initial multidataset validation with a controlled mechanism experiment. Additional health datasets with independently observed overlap, prevalence, sample size, and minority heterogeneity are required.

The strongest Vigitel factorial finding comes from one 5000-record sample, one cross-validation partition, one CTGAN configuration, and shared within-fold candidate pools. The strengthened experiment uses three repeated five-fold partitions and 30 training epochs for computational feasibility. Fold-specific generator seeds avoid reusing a single fitted generator, and repeated shuffles measure some partition variability, but all repetitions still reuse the same two source samples. More generator seeds, higher-capacity sensitivity runs, and independent datasets are necessary before formal confirmatory claims.

The fixed lambda grid was exploratory because performance and configuration selection used the same folds. The strengthened analysis fixes the separability weight in advance, but it does not estimate a clinically optimal trade-off between discrimination and minority diversity. The score also depends on a random-forest confidence term; additional boundary models and label-permutation controls are needed to test classifier-specific shortcuts.

The TVAE condition is not directly symmetric with CTGAN. SDV 1.17 does not implement native TVAE conditional sampling, so TVAE was fitted to minority-class features. This is a reproducible conditional-generation strategy, but its low diversity shows that post-selection cannot compensate for a collapsed or highly concentrated candidate pool. A diffusion model and a generator with native conditional support should be added before making a broad generator-agnostic claim.

Minority coverage and diversity are geometric proxies rather than clinically validated phenotype measures. The COVID-19 death class may contain several pathways that are not represented by unsupervised distances. Blinded clinician review and prespecified phenotype or subgroup definitions are required to determine whether removed variation is clinically meaningful. Calibration estimates are also unstable in the small COVID-19 folds.

The privacy analysis is limited to exact matches, nearest-record distances, and a membership proxy. It does not include attribute inference, shadow-model membership attacks, linkage attacks, or a formal differential-privacy guarantee. No synthetic data release should be justified from the present privacy results.

Finally, external or temporal validation is absent. Vigitel spans multiple survey years, but the current evaluation is not a prospective temporal split. The COVID-19 cohort is small and institution-specific. At least one temporal or cross-site evaluation is needed to establish robustness under distribution shift.

These limitations define the next additions to the manuscript: 2–4 further health datasets, repeated partition and generator seeds, a native conditional third generator, expert-informed minority coverage, stronger privacy attacks, and external or temporal validation. If SepAware fails under some of these conditions, those failures should be reported as operational guidance about when separability control should not be used.

7 Conclusions

Class imbalance and class overlap are not interchangeable problems. In the completed synthetic-to-real factorial experiment, explicit separability pressure explains the large Vigitel utility change but not the COVID-19 variation. In the strengthened augmentation experiment, SepAware improves CTGAN modestly on both datasets, does not improve TVAE, and is often matched or exceeded by random oversampling. Its closer minority coverage is accompanied by reduced diversity.

SepAware should therefore be understood as an auditable mechanism for controlling and studying class-conditional geometry, not as a universally superior augmentation method. Its practical use requires evidence that a predictive gain survives comparison with simple baselines and does not come from removing difficult but legitimate minority cases. The current two-dataset study establishes this conditional research direction and provides the experiment structure needed to add further datasets, seeds, generators, clinical coverage measures, and external validation.

Acknowledgments

[AUTHOR TO COMPLETE: acknowledge scientific, technical, institutional, and data-access contributions that do not meet authorship criteria.]

Generative AI assistance was used during manuscript preparation for drafting, code assistance, and language revision. The authors are responsible for verifying the analyses, citations, interpretation, and final text. This disclosure should be updated to match the exact JMIR submission policy and the authors' final use of AI tools.

Funding

[AUTHOR TO COMPLETE: identify each funder and grant or scholarship number, including support for publication charges, or state that no external funding was received.]

Conflicts of Interest

[AUTHOR TO COMPLETE: disclose financial and nonfinancial conflicts. If none, state "None declared."]

Authors' Contributions

[AUTHOR TO COMPLETE after confirming the author list. Assign CRediT roles such as conceptualization, methodology, software, formal analysis, supervision, validation, visualization, and writing.]

Data Availability

Vigitel is a publicly documented Brazilian surveillance system [23]. [AUTHOR TO COMPLETE: provide the exact access route, extract version, survey years, and terms governing the 5000-record sample.] The COVID-19 hospital-mortality dataset is not distributed with this manuscript. [AUTHOR TO COMPLETE: identify the source institution, access restrictions, and process for qualified access, if permitted.]

Code Availability

The Overleaf package includes the manuscript source, experiment protocol, strengthened analysis scripts, executed notebook entry point, fold-level results, and figure-generation code. Before submission, the authors should publish a data-safe repository snapshot, add its permanent URL and license here, and confirm that no restricted patient-level records are present.

Ethical Considerations

[AUTHOR TO COMPLETE: provide the responsible ethics committee or institutional review board, approval or exemption number, consent or waiver determination, deidentification statement, and compensation statement where applicable. These facts cannot be inferred from the local research files.]

Software Environment

The strengthened experiment used Python 3.12, NumPy 1.26.4, pandas 2.2.2, scikit-learn 1.6.1, SDV 1.17.2, CatBoost 1.2.10, XGBoost 3.2.0, imbalanced-learn 0.12.3, and statsmodels 0.14.2. Generators were run on CPU. The project-local environment pins NumPy 1.26.4 for binary compatibility with the available scientific extensions.

Abbreviations

AUPRC area under the precision-recall curve

AUROC area under the receiver operating characteristic curve

CTGAN conditional tabular generative adversarial network

HM hypothesis margin

SI separability index

SMOTE synthetic minority oversampling technique

TSTR train on synthetic, test on real

TVAE tabular variational autoencoder

References

- [1] Andre Goncalves, Priyadip Ray, Braden Soper, Jennifer Stevens, Linda Coyle, and Ana Paula Sales. Generation and evaluation of synthetic patient data. *BMC Medical Research Methodology*, 20:108, 2020. doi: 10.1186/s12874-020-00977-1.
- [2] Mikel Hernadez, Gorka Epelde, Ane Alberdi, Rodrigo Cilla, and Debbie Rankin. Synthetic tabular data evaluation in the health domain covering resemblance, utility, and privacy dimensions. *Methods of Information in Medicine*, 62(S 01):e19–e38, 2023. doi: 10.1055/s-0042-1760247.

- [3] Bayrem Kaabachi, Jérémie Despraz, Thierry Meurers, Karen Otte, Mehmed Halilovic, Bogdan Kulynych, Fabian Prasser, and Jean Louis Raisaro. A scoping review of privacy and utility metrics in medical synthetic data. *npj Digital Medicine*, 8:60, 2025. doi: 10.1038/s41746-024-01359-3.
- [4] Trivellore E. Raghunathan, Jerome P. Reiter, and Donald B. Rubin. Multiple imputation for statistical disclosure limitation. *Journal of Official Statistics*, 19(1):1–16, 2003.
- [5] Zhenchen Wang, Puja Myles, and Allan Tucker. Generating and evaluating cross-sectional synthetic electronic healthcare data: Preserving data utility and patient privacy. *Computational Intelligence*, 37(2):819–851, 2021. doi: 10.1111/coin.12427.
- [6] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems*, volume 27, pages 2672–2680, 2014.
- [7] Edward Choi, Siddharth Biswal, Bradley Malin, Jon Duke, Walter F. Stewart, and Jimeng Sun. Generating multi-label discrete patient records using generative adversarial networks. In *Proceedings of the 2nd Machine Learning for Healthcare Conference*, volume 68 of *Proceedings of Machine Learning Research*, pages 286–305, 2017. URL <https://proceedings.mlr.press/v68/choi17a.html>.
- [8] Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [9] Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations*, 2014.
- [10] Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. TabDDPM: Modelling tabular data with diffusion models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 17564–17579, 2023. URL <https://proceedings.mlr.press/v202/kotelnikov23a.html>.
- [11] Vasileios C. Pezoulas, Dimitrios I. Zaridis, Eugenia Mylona, Christos Androutsos, Kosmas Apostolidis, Nikolaos S. Tachos, and Dimitrios I. Fotiadis. Synthetic data generation methods in healthcare: A review on open-source tools and methods. *Computational and Structural Biotechnology Journal*, 23:2892–2910, 2024. doi: 10.1016/j.csbj.2024.07.005.
- [12] Mikel Hernandez, Pablo A. Osorio-Marulanda, Mikel Catalina, Lorea Loinaz, Gorka Epelde, and Naiara Aginako. Comprehensive evaluation framework for synthetic tabular data in health: Fidelity, utility and privacy analysis of generative models with and without privacy guarantees. *Frontiers in Digital Health*, 7:1576290, 2025. doi: 10.3389/fdgth.2025.1576290.
- [13] Ivana Nanevski, Maryam Mohebi, Sebastian J"ager, Karen Otte, Fabian Prasser, Matthias Schulte-Althoff, Daniel F"urstenau, and Felix Biessmann. Evaluating the quality of tabular synthetic data in health care. *PLOS Digital Health*, 5(7):e0001522, 2026. doi: 10.1371/journal.pdig.0001522.
- [14] Ghadeer O. Ghosheh, Tingting Zhu, and Jin Li. A survey of generative adversarial networks for synthesizing structured electronic health records. *ACM Computing Surveys*, 56(6), 2024. doi: 10.1145/3636424.

- [15] Takaya Saito and Marc Rehmsmeier. The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3): e0118432, 2015. doi: 10.1371/journal.pone.0118432.
- [16] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16: 321–357, 2002. doi: 10.1613/jair.953.
- [17] Haibo He and Edwardo A. Garcia. Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, 21(9):1263–1284, 2009. doi: 10.1109/TKDE.2008.239.
- [18] Margerie Huet-Dastarac, Fida K. Dankar, Dan Liu, Samer El Kababji, Lisa Pilgram, and Khaled El Emam. An evaluation of pretrained generative models for augmenting small health data: Comparative modeling study. *Journal of Medical Internet Research*, 28:e88678, 2026. doi: 10.2196/88678.
- [19] Andrew Yale, Saloni Dash, Ritik Dutta, Isabelle Guyon, Adrien Pavao, and Kristin P. Bennett. Assessing privacy and quality of synthetic health data. In *Proceedings of the Conference on Artificial Intelligence for Data Discovery and Reuse*, pages 1–4, 2019. doi: 10.1145/3359115.3359124.
- [20] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography Conference*, pages 265–284, 2006. doi: 10.1007/11681878_14.
- [21] Hansle Gwon, Imjin Ahn, Yunha Kim, Hee Jun Kang, Hyeram Seo, Heejung Choi, Ha Na Cho, Minkyong Kim, JiYe Han, Gaeun Kee, Seohyun Park, Kye Hwa Lee, Tae Joon Jun, and Young-Hak Kim. LDP-GAN: Generative adversarial networks with local differential privacy for patient medical records synthesis. *Computers in Biology and Medicine*, 168:107738, 2024. doi: 10.1016/j.combiomed.2023.107738.
- [22] A. T. Christoforou, S. K. B. Spohn, T. Sprave, T. Fechter, A. Ruehle, N. H. Nicolay, I. Popp, A. L. Grosu, J. C. Peeken, A. H. Thieme, T. Stylianopoulos, I. Strouthos, K. Ferentinos, Y. Roussakis, and C. Zamboglou. A framework to create, evaluate and select synthetic datasets for survival prediction in oncology. *Computers in Biology and Medicine*, 192:110198, 2025. doi: 10.1016/j.combiomed.2025.110198.
- [23] Regina Tomie Ivata Bernal, Betine Pinto Moehleck Iser, Deborah Carvalho Malta, and Rafael Moreira Claro. Sistema de vigilancia de fatores de risco e protecao para doencas cronicas por inquerito telefonico (Vigitel): Mudanca na metodologia de ponderacao. *Epidemiologia e Servicos de Saude*, 26(4):701–712, 2017. doi: 10.5123/S1679-49742017000400003.
- [24] Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: Unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, pages 6638–6648, 2018.
- [25] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.
- [26] Noah Hollmann, Samuel Muller, Lennart Purucker, Arjun Krishnakumar, Max K"orfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small

A Reproducibility and Artefact Status

Table 9 lists the status of every research artefact consulted while preparing this paper and states how each was treated. Superseded, protocol-only, or failed runs are excluded from all numerical claims in the main text; only executed, leakage-controlled five-fold notebooks and their exported CSV tables are reported as evidence.

Table 9: Status of research artefacts consulted during manuscript preparation.

Artefact	Status	Manuscript treatment
Older dataset-specific fixed- λ notebooks	Superseded single-split studies	Excluded from numerical claims because they use one 75/25 split.
Combined SEPAWARE fixed- λ notebook	Executed five-fold study	Authority for fold-level fixed- λ comparisons (Table 3).
Protocol-audited 2 ² cross-validation notebook	Executed	Superseded by the replicated factorial notebook below; retained only for earlier condition summaries and correlation interpretation during development.
Replicated factorial ANOVA notebook (with results)	Latest executed experiment	Authority for the 40 fold responses, ANOVA, effects, and diagnostics (Section 4.3).
Factorial assumption-test script and CSVs	Executed after the notebook above	Authority for expanded residual tests and the blocked-versus-unblocked sensitivity analysis.
COVID DACPF raw and summary CSVs	Executed	Authority for augmentation results in Table 2.
TabPFN augmentation notebook	Failed dependency, no output	Protocol only; no result claimed.
Unexecuted causal-filter notebooks	Incomplete protocol	Discussed as development history / future work, not evidence.
Cluster-then-augment notebook	No authoritative exported result located	Excluded from numerical results.
Strengthened two-dataset augmentation script	Executed three-repeat five-fold experiment	Authority for conventional baselines, CTGAN/TVAE augmentation, calibration, minority coverage/diversity, and privacy-proxy results.
Controlled prevalence-by-overlap script	Executed mechanism experiment	Authority for the independent prevalence and overlap manipulation; not counted as a third health dataset.

A.1 Result-authority decisions

The rerun factorial experiment differs numerically from earlier protocol-audited notes because CT-GAN is stochastic even under a fixed, recorded software stack; this paper uses the newly executed observations consistently and does not mix values across reruns. An unblocked 2^2r decomposition that treats folds as replications is also present in the repository and assigns COVID-19 90.01% error and Vigitel 16.92% error; the main text instead uses the blocked model (Equation 2), which separates fold variation from treatment variation and yields 81.53% and 3.59% respectively (Table 6). Both calculations exist in the repository, but they answer different questions, and only the blocked model is used for the paper’s conclusions.

The qualification factorial protocol evaluates standalone synthetic datasets (TSTR). The strengthened experiment separately reports real-plus-synthetic augmentation and does not pool the two regimes. AUPRC, calibration, minority geometry, exact-match, and membership-proxy measures are reported only for the strengthened run. Attribute inference, formal privacy guarantees, and external validation remain absent and are reported as limitations rather than estimated after the fact.